

# Machine Learning Prediction and AI Modeling of Music Genre Perception Based on fMRI Data

## Abstract

**Keywords:** *Machine Learning, auditory spectrotemporal modeling, fMRI analysis, AI*

## I Introduction

Functional magnetic resonance imaging (fMRI) captures brain activity through localized oxygenization changes, a characteristic referred to as Blood Oxygenization Level Dependent (BOLD) signal [1, 2, 3]. When neurons in a certain brain region are active, they require more oxygen and such increased neural activity is followed by a slight delayed increase in blood flow, a hemodynamic response that can be captured to peak several seconds after the stimulus [2]. The BOLD signal is hence a proxy to map brain activation and understand the correspondence between stimulus and brain response. However, the delay and instrumental resolution limitation also poses difficulty to track the temporal information evolution.

One important consideration in MRI research is to tame the variability of cortical organization across different individuals. Variability in brain anatomy, vascular structure, functional localization requires advanced and robust registration techniques to harmonize these discrepancies [1]. However, many open-source tools and consortium have tackled this problem [4, 5, 6]. fMRI remains an invaluable tool for non-invasive characterization of the structural, functional, and physiological properties of the human brain, particularly through the integration with high spatial resolution structural MRI [7].

Music perception is a premier example illustrating the complex translation of raw input (analogue sound-waves) into complex auditory brain wave signals. The latter traverse the intricate brain networks and can be probed through fMRI with advanced statistical computing, machine learning (ML), analytics, and

artificial intelligence (AI) techniques [8, 9, 10]. Sound perception begins when inner hair cells in the cochlea transduce sound waves through mechanical vibrations into biological (electro-chemical) neural signals. Relayed in the midbrain and thalamus (specifically the medial geniculate nucleus), these multispectrum waves ascend through the auditory pathway before reaching the primary auditory cortex. Brain networks of higher-order regions encode (analyze) and decode (synthesize) complex auditory features, such as pitch, timbre, and rhythm [11].

The human perception of music begins with the hierarchical encoding of acoustic features within higher-order auditory cortices. While the primary auditory cortex in Heschl’s gyrus performs initial analysis of frequency and intensity, adjacent regions in the superior temporal gyrus (STG) are critical for parsing more complex musical elements. There is functional dissociation where the posterior STG and planum temporale are preferentially engaged in analyzing spectrotemporal patterns and the timing of sound events that are crucial for rhythm perception. In contrast, the anterior STG is more involved in processing spectral information (frequency patterns). It contributes to the perception of pitch and timbre and creates a “what” pathway for auditory object recognition. The human neural architecture acts as a sophisticated Fourier analyzer/synthesizer engine, transforming the analogue auditory signal into parallel streams of information representing pitch contours, rhythmic patterns, and timbral qualities, which form the building blocks of musical texture and genre [12].

The continuous brain encoding and decoding of temporal auditory signals maps these separately analyzed features into a coherent musical experience through large-scale neural networks that extend beyond the auditory cortex. The ventral stream, connecting the anterior STG with the inferior frontal gyrus (IFG) and other regions of the temporal lobe, is crucial for identifying musical patterns and tonal syntax, leading to both understanding and synthesizing of melodic signals. The IFG, in particular, is implicated in processing structural regularities and violations of expectation in melody and harmony. Concurrently, a dorsal stream connects the posterior STG with parietal and premotor cortices. This pathway is vital for sensorimotor integration, mapping sound to action (as in tapping along to a beat), tracking and anticipating (forecasting) the temporal unfolding of music, which is fundamental to our perception of rhythm

and meter [13]. The constant interactions between these streams allows the brain to synthesize disparate features into a unified and dynamically evolving musical percept.

Ultimately, the recognition of abstract musical concepts such as *texture*, e.g., polyphony vs. monophony, and *genre* relies on the integration of these synthesized musical percepts with cognitive systems for memory, emotion, and categorization. The perception of genre is not merely an acoustic analysis but a categorical quantification engaging the medial prefrontal cortex (mPFC) and hippocampus, and accessing learned stylistic schemas and long-term memories associated with past listening experiences. The frontoparietal control network is engaged to manage attentional resources and working memory, which are necessary to follow the interplay of different melodic lines within a complex texture. For instance, limbic areas like the amygdala and nucleus accumbens track the emotional valence and reward value of the music, which are strongly tied to genre-specific conventions. These processes affect the entire perceptual experience and solidify the synthesis of sound into meaning [14].

The main contributions of this study include .....

... This article is organized as follows. In section II, we review previous work and relevant studies. In section ....

## II Previous Studies

Modern imaging tools such as high spatial resolution fMRI, MEG/EEG, and intracranial recordings along with computational machine learning techniques have assisted neuroimaging researchers to gain deeper understanding of how our brain process sound, speech and music over the past few decades [15, 16, 17, 18, 19]. Key findings point to **topographic maps** in auditory cortex for acoustic features, specialized **pitch-processing regions**, and distinct neural populations that track **tonal structure and melody**. We summarize some key findings as below.

**Cortex Representation of Music:** Basic acoustic features derived from music signal are mapped onto the auditory cortex topographically and systematically. One well-known results is that the Heschl's gyrus is tonotopically organized[20]. This tonopoic organization hypothesis is first observed in invasive animal

studies including: cats [21, 22], dogs[23], and grey squirrels [24]. These studies showed an ordered representation of the cochlear basilar partition in the auditory region of the cerebral cortex in animals. The noninvasive techniques to confirm equivalent tonotopic organization was then later carried out in humans through PET [25], MEG [26] and fMRI studies[20]. More recently, with recent advances in high spatial resolution fMRI, it is demonstrated that distinct subregions of human auditory cortex have different range of spectrotemporal modulation rates [18]. This work reveals that human auditory processing mechanism are finetuned to most behaviorally relevant sounds. Another recent study uncovers striking **hemispheric asymmetries** demonstrating the left and right auditory cortices show differential sensitivity to temporal vs spectral music information [27, 15, 28]. Albouy et al [15, 27] used filtered a cappella songs to independently manipulate spectral and temporal information, revealing a double dissociation in behavior and brain responses. Namely, that **speech content was decoded exclusively by left auditory cortex, whereas melodic content was processed in right auditory cortex**. Such findings unify longstanding theories of auditory lateralization within a spectrotemporal modulation framework[28].

**Higher-order auditory cortex** Beyond primary auditory areas, higher-order auditory cortex exhibits categorical organization for complex sound processing including music. Norman-Haignère et al. [12] applied voxel decomposition to fMRI responses to natural sounds and discovered distinct cortical response components, revealing a music-selective component in non-primary auditory cortex (anterior STG) independent of low-level acoustics and spatially distinct from speech-selective regions. These signals, obscured in raw fMRI data, emerged only through decomposition, suggesting overlapping neural populations (within voxels). A follow-up study combined electrocorticography (ECoG) and fMRI to further resolve these populations: Norman-Haignère [29] specifically identified a neural population in human auditory cortex selective for song (music with vocals), spatially intermingled with but separable from instrumental-music and speech representations.

**Pitch Representation in the Cortex: “Absolute” vs. “Relative” Pitch** Pitch is a fundamental building block of melody and harmony that characterize perceived highness (or lowness) of a sound. Neuroscience studies indicate that pitch is represented in specialized cortical regions and encodes multiple forms: both as absolute pitch (specific frequency values) and as relative pitch (relative intervals or relationships between notes). Using high-field fMRI, De Angelis et al. pinpointed a pitch-processing center in the human auditory cortex for natural sounds [19]. They built an encoding model that estimated each sound’s perceived pitch height (function of fundamental frequency) and pitch salience (related to harmonic structure). Their model’s predictions identified an anterolateral Heschl’s gyrus (alHG) and adjacent superior temporal gyrus region that encoded pitch. In contrast, relative pitch, the perception of music note intervals (and constituent for higher order feature like melody) independent of absolute frequency is found to engage broader cortical networks. Humans easily recognize a familiar melody played in a different key, reflecting a translational (transposition) invariance characterizing the relative pitch processing in brain. Han et al [30] conducted an fMRI event-related adaptation and cross-classification experiment to find regions tolerant to melody transposition. They discovered significant adaptation elicited in right STG and left mid-temporal gyrus when a melody was repeated at a different pitch level, indicating these auditory regions treat transposed melodies as the “same” melody.

Spatial segregation of pitch-related computations is supported by electrocortigraphy. Sankaran et al. [17] identified distinct superior temporal gyrus (STG) sites encoding (i) absolute pitch, (ii) intervals (relative pitch changes), and (iii) tonal expectations (given prior melodic context). Notably, sites encoding tonal expectations were music-selective, responding to harmonic structure in melodies but not analogous patterns in speech, whereas absolute pitch and interval regions exhibited domain-general responses, processing both music and speech prosody. This double dissociation highlights specialized neural populations: some compute low-level acoustic features shared across domains, while others are dedicated to music-specific tonal hierarchies. Earlier fMRI work by Casey [8] corroborates this division: absolute pitch (shared with acoustic chromagram features) activated overlapping networks in multiple regions lateralized to the right

such as Heschl gyrus, middle temporal gyrus, frontal pole and planum temporale etc, while relative pitch engaged focused (concentrated) clusters in the left planum polare and right planum temporale, suggesting tonal relationships are processed in higher-order auditory zones. Similarly, Janata et al. [16] identified rostral prefrontal regions tracking tonal key via a geometric “pitch torus” model, linking melodic context to spatial representations. Foster & Zatorre’s work [31] identified right Heschl’s sulcus and Intraparietal sulcus IPS activation during transposed melody comparison suggests role in relative pitch processing via auditory working memory.

**Brain Encoding and Decoding studies.** One ultimate goal for auditory neuroscience is to establish a mapping between sound stimuli and brain response. The *encoding phase* attempts to predict brain activity from features derived from the sound, while the *decoding phase* aims to reconstruct features of music from brain activity. In the encoding phase, one fits neural cortical responses from the raw music features (simple low-level spectrogram and high-level complex statistics such as tonality or genre). Casey et al. [8] used spherical searchlight regression predicting neural fMRI response patterns from music features resembling melody and harmony. Applying this model to the whole brain, they concluded that distinct music features are encoded and distributed over anatomically overlapping regions. In this brain mapping process, widespread areas (in temporal, frontal, parietal, occipital cortex) encode acoustic and absolute-pitch features while more local areas encode tonal relative-pitch features [8, Table 2]. Encoding establishes correlation between music features and brain structures. De Angelis et al.[19] identified a pitch-coding region in the anterolateral Heschl gyrus (HG) and the adjacent superior temporal gyrus (STG). Kell et al. [32] optimized a deep neural network (DNN) for music and speech recognition, and indicated that their trained DNN mimicks the brain’s hierarchical organization along the auditory pathway, validating the encoding power of learned features in a DNN.

On the other hand, *decoding* focuses on the music aspects that can be read out from brain activity. Santoro et al. [18] demonstrated a form of decoding by reconstructing the spectrotemporal modulation content of heard sounds from fMRI signals, i.e., they “reverse-engineered” the brain’s sound processing by linking fMRI activity patterns to specific modulation properties. In the domain of music, Madsen

et al. [33] used EEG to study music listener attention span and found engagement tends to decrease across repeated exposures for familiar ones, but that unfamiliar musical styles can sustain an audience’s interest, in particular in individuals with previous music training. Casey [8] used Multivariate Pattern Analysis to decode which song was being heard from the fMRI voxel patterns in auditory cortex, achieving significant identification accuracy in a 5-class (ambient, country, metal, rock-and-roll, and symphony) genre classification for the 25 stimuli. There are emerging recent end to end decoding examples. One striking recent example of decoding is a 2023 study where researchers used intracranial EEG from neurosurgery patients to reconstruct portions of the Pink Floyd song “Another Brick in the Wall” that the patients were listening to [34]. In another recent work, Brain2music establishes a music language model [35] that can auto-generate synthetic music from brain activity.

**Music perception with speech and sound** Music and speech processing both involve hierarchical structure (notes or phonemes combining into phrases or melodies), and thus the brain’s strategies for abstraction and prediction may be similar. [17]

**Individual and confounding factors** Individual neural circuits wiring (musicians and average non-trained individuals)...Albouy [27, 15] also showed that top-down factors (e.g. attention, training) can modulate these asymmetries

### III Methods

We begin by presenting the music genre fMRI data and the corresponding data pre-processing.

#### III.1 Dataset Description

We utilize the fMRI music genre dataset provided in [36]. The dataset constitutes 12 training runs and 6 test runs, each run constitutes 41 music clips with a total of 492 training samples and 246 testing samples. The same *experimental condition* is exposed to different subjects in each run indices. In other words, all 5 participants were exposed to the *same* event-related fMRI study design, i.e., all participants undergo the same fMRI runs with the same musical stimuli. For a fixed individual, the training runs and testing runs

are independent from each other. For the *training runs*, the music genre stimulus order is random and the selection of the 15-second music clip from the GTZAN dataset [37] is randomly clipped. While *testing runs* were more structured and the music clips played for each genre are the same. Furthermore, in the testing run, the order of exposure follows a cyclic pattern. For example, in testing run 1 the order of exposure is always: pop  $\rightarrow$  rock  $\rightarrow$  country  $\rightarrow$  blues  $\rightarrow$  disco  $\rightarrow$  metal  $\rightarrow$  reggae  $\rightarrow$  classical  $\rightarrow$  hiphop  $\rightarrow$  jazz  $\rightarrow$  pop  $\rightarrow \dots$ . The music track selected from the GTZAN dataset are exactly identical within the same run. However, comparing different test runs, the music clip selected are not the same. The same cyclic experimental setup is preserved nevertheless. The authors omitted the first training scan for each run in their analysis [9]. The original dataset has a size of 43.33GB, which can be accessed via openneuro[36]. The original data dimension is  $96 \times 96 \times 68 \times 410$  with 18 runs for each of the 5 individuals. After applying the fMRIPrep procedure and mapping to MNINLin2009cAsym space the data dimension is  $97 \times 115 \times 97 \times 410$ , the total preprocessed data size (including all metadata) is 124.89GB. After selecting all the relevant voxels in the right superior temporal gyrus (Region 66 from Desikan Atlas), 2105 voxels are selected (See Figure 1). The effective data size that we use for ML is 0.58 GB, where for a single individual the training data size is (492, 2105, 10) and the testing data size is (246, 2105, 10) after splitting the 410 timesteps in one run into 41 samples.

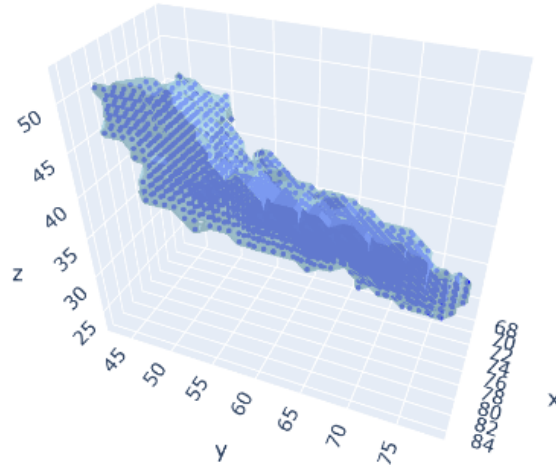


Figure 1: The shape of Right Superior Temporal Gyrus in the MNI152Nlin2009cAsym space.

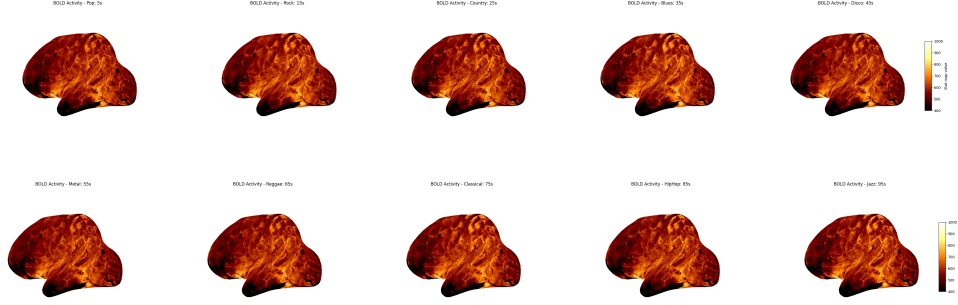


Figure 2: Timesteps for the cortex activation at T=5 (Pop),15 (Rock),25 (Country),35 (Blues),45 (Disco), 55(Metal), 65(Reggae),75 (Classical), 85 (Hip-hop), 95(Jazz). (The indices start at 0).

### III.2 Visualizations on the music stimulus

We'll present several alternative ways to visually characterize the music textures of the music stimulus exposed in the experiment.

... describe all figures /verbalize what is shown ....

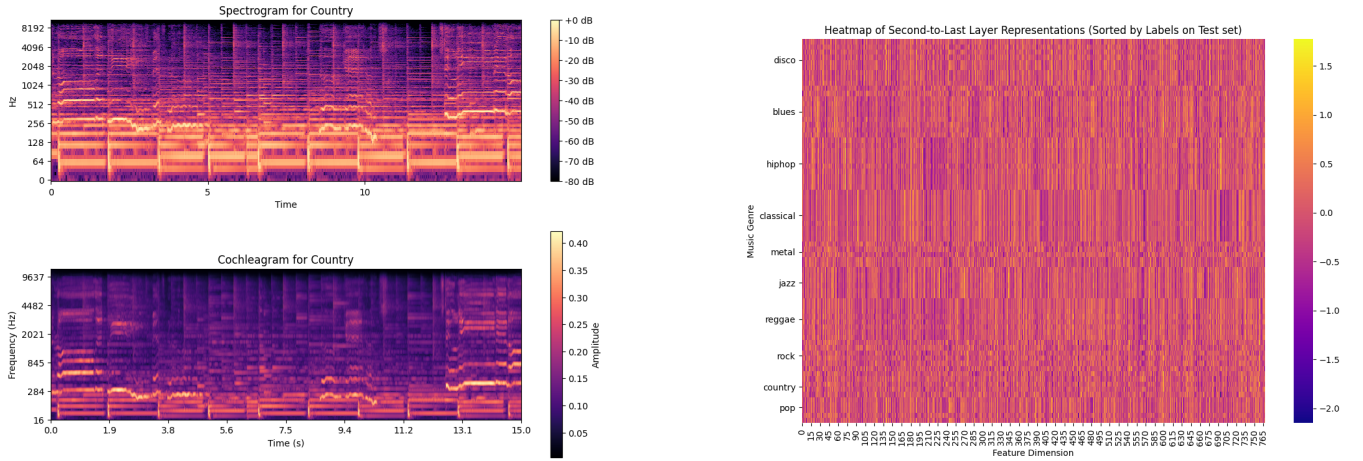


Figure 3: (Left) Ways for extracting spectrotemporal information from the music input of the first country music clip in Training Run 1 through absolute pitch. The top is a spectrogram generated from librosa and the bottom is a cochleagram generated from [38]. (Right) Homologies captured from finetuned pretrained speech transformer model (Distil-Hubert [39]) visualized on unseen music clip test set not seen in training

### III.3 Dataset Preprocessing

fMRI packages: fMRIPrep (Integrates Brain extraction (or “skull stripping”), motion correction, slice timing correction, smoothing, temporal filtering, registration, and normalization. ), SPM

Why subject 4 is particularly tricky to deal with

Results included in this manuscript come from preprocessing performed using *fMRIPrep* 21.0.1 ([40]; [41, RRID:SCR\_016216] ), which is based on *Nipype* 1.6.1 ([42]; [43, RRID:SCR\_002502] ).

**Anatomical data preprocessing** A total of 1 T1-weighted (T1w) images were found within the input BIDS dataset. The T1-weighted (T1w) image was corrected for intensity non-uniformity (INU) with *N4BiasFieldCorrection* [44], distributed with ANTs 2.3.3 [45, RRID:SCR\_004757], and used as T1w-reference throughout the workflow. The T1w-reference was then skull-stripped with a *Nipype* implementation of the *antsBrainExtraction.sh* workflow (from ANTs), using OASIS30ANTs as target template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using *fast* [FSL 6.0.5.1:57b01774, RRID:SCR\_002823, 46]. Brain surfaces were reconstructed using *recon-all* [FreeSurfer 6.0.1, RRID:SCR\_001847, 47], and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray-matter of Mindboggle [RRID:SCR\_002438, 48]. Volume-based spatial normalization to one standard space (MNI152NLin2009cAsym) was performed through nonlinear registration with *antsRegistration* (ANTs 2.3.3), using brain-extracted versions of both T1w reference and the T1w template. The following template was selected for spatial normalization: *ICBM 152 Nonlinear Asymmetrical template version 2009c* ([49, RRID:SCR\_008796]; TemplateFlow ID: MNI152NLin2009cAsym).

**Functional data preprocessing** For each of the 18 BOLD runs found per subject (across all tasks and sessions), the following preprocessing was performed. First, a reference volume and its skull-stripped version were generated using a custom methodology of *fMRIPrep*. Head-motion parameters with respect to the BOLD reference (transformation matrices, and six corresponding rotation and translation param-

eters) are estimated before any spatiotemporal filtering using `mcflirt` [FSL 6.0.5.1:57b01774, 50]. The BOLD time-series (including slice-timing correction when applied) were resampled onto their original, native space by applying the transforms to correct for head-motion. These resampled BOLD time-series will be referred to as *preprocessed BOLD in original space*, or just *preprocessed BOLD*. The BOLD reference was then co-registered to the T1w reference using `bbregister` (FreeSurfer) which implements boundary-based registration [51]. Co-registration was configured with six degrees of freedom. Several confounding time-series were calculated based on the *preprocessed BOLD*: framewise displacement (FD), DVARS and three region-wise global signals. FD was computed using two formulations following Power (absolute sum of relative motions, [52]) and Jenkinson (relative root mean square displacement between affines, [50]). FD and DVARS are calculated for each functional run, both using their implementations in *Nipype* [following the definitions by 52]. The three global signals are extracted within the CSF, the WM, and the whole-brain masks. Additionally, a set of physiological regressors were extracted to allow for component-based noise correction [*CompCor*, 53]. Principal components are estimated after high-pass filtering the *preprocessed BOLD* time-series (using a discrete cosine filter with 128s cut-off) for the two *CompCor* variants: temporal (tCompCor) and anatomical (aCompCor). tCompCor components are then calculated from the top 2% variable voxels within the brain mask. For aCompCor, three probabilistic masks (CSF, WM and combined CSF+WM) are generated in anatomical space. The implementation differs from that of Behzadi et al. in that instead of eroding the masks by 2 pixels on BOLD space, the aCompCor masks are subtracted a mask of pixels that likely contain a volume fraction of GM. This mask is obtained by dilating a GM mask extracted from the FreeSurfer’s *aseg* segmentation, and it ensures components are not extracted from voxels containing a minimal fraction of GM. Finally, these masks are resampled into BOLD space and binarized by thresholding at 0.99 (as in the original implementation). Components are also calculated separately within the WM and CSF masks. For each CompCor decomposition, the  $k$  components with the largest singular values are retained, such that the retained components’ time series are sufficient to explain 50 percent of variance across the nuisance mask (CSF, WM, combined, or temporal). The remaining components are dropped from consideration. The head-motion estimates calculated in the correction step were also placed

within the corresponding confounds file. The confound time series derived from head motion estimates and global signals were expanded with the inclusion of temporal derivatives and quadratic terms for each [54]. Frames that exceeded a threshold of 0.5 mm FD or 1.5 standardised DVARS were annotated as motion outliers. The BOLD time-series were resampled into standard space, generating a *preprocessed BOLD run in MNI152NLin2009cAsym space*. First, a reference volume and its skull-stripped version were generated using a custom methodology of *fMRIPrep*. The BOLD time-series were resampled onto the following surfaces (FreeSurfer reconstruction nomenclature): *fsnative*, *fsaverage5*. All resamplings can be performed with *a single interpolation step* by composing all the pertinent transformations (i.e. head-motion transform matrices, susceptibility distortion correction when available, and co-registrations to anatomical and output spaces). Gridded (volumetric) resamplings were performed using `antsApplyTransforms` (ANTs), configured with Lanczos interpolation to minimize the smoothing effects of other kernels [55]. Non-gridded (surface) resamplings were performed using `mri_vol2surf` (FreeSurfer).

Many internal operations of *fMRIPrep* use *Nilearn* 0.8.1 [56, RRID:SCR\_001362], mostly within the functional processing workflow. For more details of the pipeline, see the section corresponding to workflows in *fMRIPrep*'s documentation.

### III.4 Machine Learning Prediction and AI Modeling

...Let's include a technical description of the exact models we used to analyze the raw audio signals (spectrograms), the fMRI brain imaging data, and spacekime analytics (if we can get these to work well)....

## IV Results

### IV.1 Individual tests

### IV.2 Intersubject Generalisation

Cortical surface analysis

left and right sphere

neuroparc

### IV.3 Cross ROI study (Different Brain Atlas)

Destrieux: High-resolution cortical parcellation for detailed surface-based studies.

Desikan: Balanced, widely adopted for cortical analyses in FreeSurfer.

AAL: Volume-based, includes subcortical regions, suited for MNI-standardized studies.

Different asymmetric spaces, MNI152NLin6 and MNI152NLibn2009c

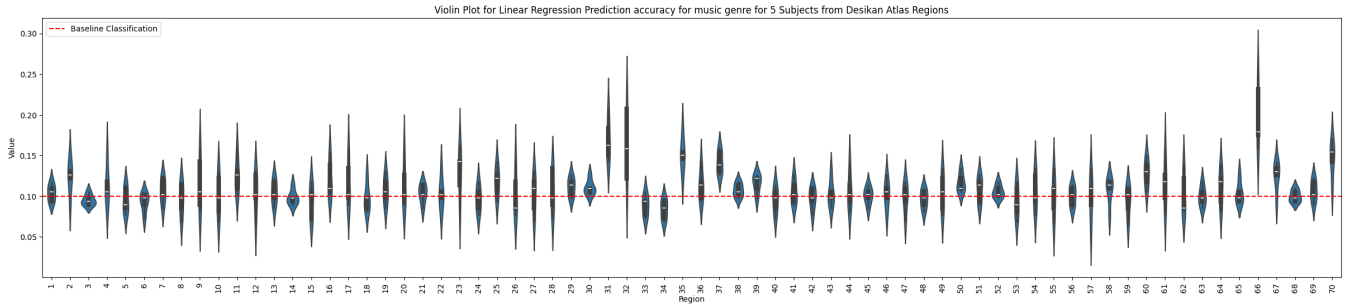


Figure 4: Different ROIs parcellated from Desikan atlas where the voxels are regressed on the training and statistics collected on the test set. Region-31 is Left Superior Temporal Gyrus. Region-32 is Left Supramarginal Gyrus. Region-35 is Left Transverse Temporal Cortex. Region-66 is right superior temporal Gyrus. Region-70 is the right transverse temporal cortex.

Put in the spatial segmentation of these regions here... (Sagittal; Axial; Coronal)

### IV.4 Machine Learning Model runs on Desikan Atlas Right superior temporal Gyrus

#### IV.4.1 MUSIC GENRE CLASSIFICATION BASELINES [57]

We utilize the music classification model [57] replacing the input with the specific 15s music that the patient exposed in the timespan. The raw music signal spans 15 seconds and we split into training, validation, testing set according to 8:1:1 ratio as in the dataset. We extract the abstract pitch and frequency features using melogram using the librosa library [58], power transform is applied to melogram for stability in the

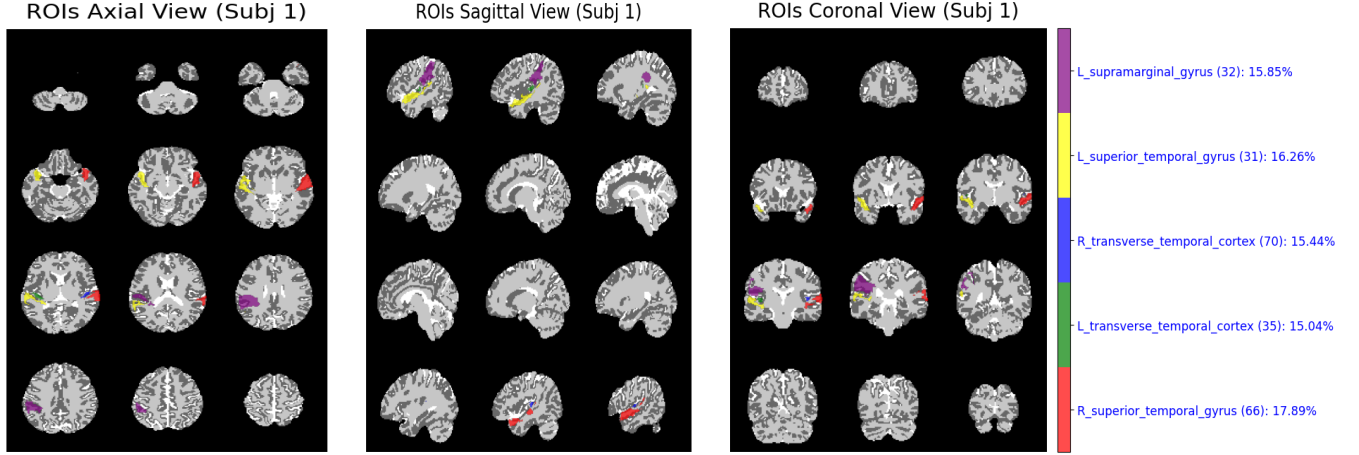


Figure 5: The spatial visualizations as 2D slices from different views, the color labeled regions for subject 1 from raw linear regression is also indicated as colormap. The ROI segmentation is transformed to MNI152NLin2009cAsym space. Only subject 1 is shown, but the ROIs are the same for different individual in the standard atlas space.

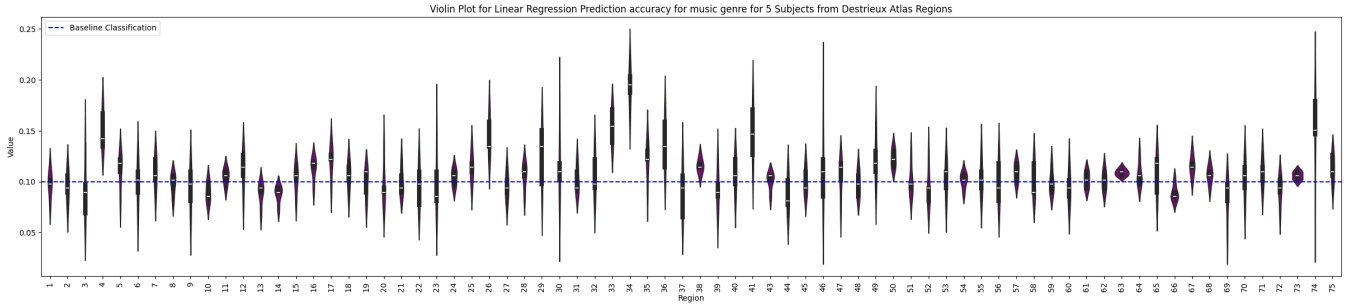


Figure 6: Different ROIs parcellated from Destrieux atlas where the voxels are regressed on the training and statistics collected on the test set. Region-4 is Subcentral gyrus and sulci. Region-26 is Supramarginal gyrus. Region-33 Anterior transverse temporal gyrus. Region-34 is Superior temporal Gyrus lateral aspect. Region-74 is superior temporal sulcus.

model training. Each raw signal feature contains 330750 data points, and the converted data for the whole data is (738, 646, 128).

#### IV.4.2 PREDICTING THE GENRE USING BRAIN SPATIOTEMPORAL STATE TENSOR

We run kNN, CNN, and an “invariant”+classification framework on the raw runs for individual 1. The problem of classifying brain state characterized by the fMRI bold signal  $\mathbb{R}^{96 \times 96 \times 68 \times 10} \rightarrow \{1, 2, \dots, 10\}$ , where  $96 \times 96 \times 68$  corresponds to the spatial dimensions and the 10 corresponds to the discrete time sampling points, and the range is indicated by the 10 music genres.

Table 1: Performance Comparison of Different Models

| Model                      | Accuracy (%) | Std. Deviation (%) |
|----------------------------|--------------|--------------------|
| Linear Regression          | 19.8         |                    |
| Temporal GNN               | XX.X         | X.X                |
| GNN+LSTM                   | XX.X         | X.X                |
| GRU                        | 25.2         | X.X                |
| LSTM                       | 24.4         | X.X                |
| RCNN+LSTM                  | XX.X         | X.X                |
| GRU with spatioencoding    | XX.X         | X.X                |
| GRU with 1DCNN-features    |              |                    |
| Spatial Features with LSTM | 28.6         |                    |

## V Conclusion

...to be completed ...

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